

Mnemosure

**An AI memory that says "*I don't know*" when it doesn't,
cites its source when it does,
and corrects itself when the facts change.**

Qwen Cloud Global AI Hackathon · Track 1 — MemoryAgent

github.com/jsiksn/mnemosure · pypi.org/project/mnemosure

The problem — forgetting and hallucination **feed each other**

It forgets

Decisions that *were* made quietly vanish. Summary-style memory keeps the current value but drops **why** it was decided — the reasons and history go first.

It makes things up

Decisions that were *never* made get stated anyway. Stale facts are asserted **with full confidence** long after they were overturned.

The less it remembers, the more it fills the gaps by inventing — so most "memory" add-ons make both failures **worse at the same time**.

Core idea — a memory with **three duties**

It does not invent what it cannot remember, and it does not drop what it remembers.

Store

Keep only what will matter later — decisions, changes, failures, facts. Throw away the chatter.

Link

A new decision marks the old one **superseded** ; its **reason** links back to the failure that caused it (**because**). Failures are kept forever.

Recall

Answer **grounded only in evidence**, with sources. Correct stale facts. No evidence → "**not in the record.**"

Every answer carries a confidence level — **certain / vague / unknown** — and cites the memories it used.

Same question, different behavior — real demo output

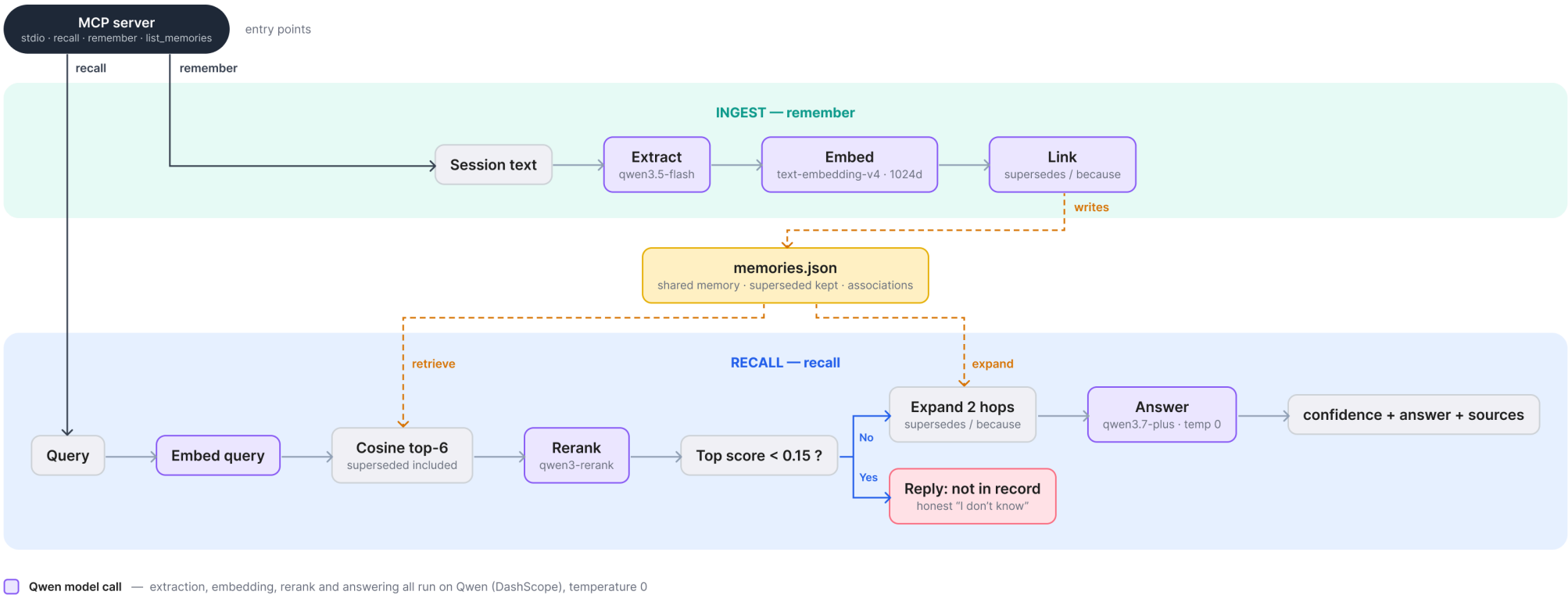
The question	A baseline answers	Mnemosure answers
"Free plan is still capped at 3 projects, right?"	hallucination — naive RAG: "Yes, 3." (overturned weeks ago)	corrects — "No, it's 1 now — cut after free-tier abuse was found." + sources
"Why did we lower the Pro price?"	omission — handoff notes: "I don't know." (the reason was dropped)	explains — "Feedback said we were pricier than competitors — so \$12 → \$9/mo." + sources
"What's the student discount?" (never discussed in any session)	handoff & naive RAG: "I don't know." — honest here too	honest — "Not in the record." — with an explicit confidence label: unknown

Verbatim behavior from the committed demo snapshot (`data/scenarios/pricing/results.json`), translated from the Korean demo. Rows 1–2 quote the baseline that failed on that question — full scores per system are on the Results slide. Answers are produced by the pipeline, never hardcoded.

Architecture — all model calls run on Qwen

Mnemosure

Source-grounded memory for AI agents — cites its sources, says “I don’t know”, corrects superseded facts.



Extract `qwen3.5-flash` · Embed `text-embedding-v4` (1024-dim) · Rerank `qwen3-rerank` · Answer `qwen3.7-plus` — DashScope, temperature 0 (deterministic).

How it **corrects** — and how it stays **honest**

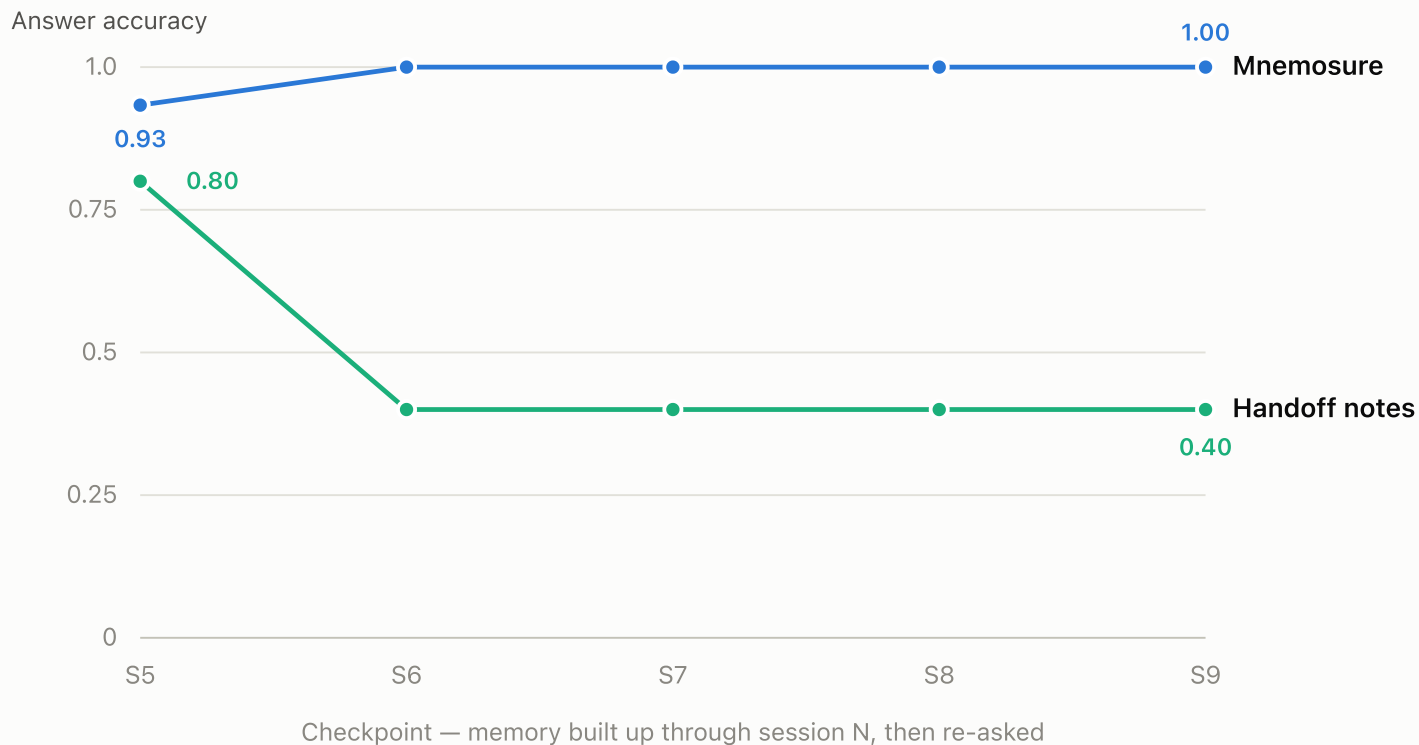
- **It looks for outdated memories on purpose.** You can only say "*it's \$9 now*" if you first find the old "*Pro is \$12*" memory. So retrieval never hides replaced facts — it brings them back together with what replaced them.
- **If the evidence is weak, it doesn't answer.** When even the best-matching memory falls below a relevance bar, the answer is "**not in the record.**" Refusing to guess is a feature, not a failure.
- **Memories are linked by judgment, not similarity.** Two memories aren't connected just because they *sound* alike — a model decides whether one actually *replaces* the other, or *explains why* it changed.
- **Every sentence stays inside the evidence.** The answer may only use the memories it retrieved, and each claim carries its source.

Results — measured on committed snapshots

Scenario	Mnemosure	Handoff notes	Naive RAG
Pricing revamp (11 questions)	11/11	4/11	4/11
Trading bot (8 questions)	8/8	2/8	2/8

- The handoff baseline keeps current values but **drops the reasons** — it misses every "why did we...?" question
- Naive RAG asserts **stale numbers** as if they were still true — hallucinations
- Baselines answer with the **same Qwen brain model** — the difference is memory design, not model horsepower
- Every question, answer and source conversation is browsable in the demo — the memories are *extracted*, not hardcoded

Why the gap keeps widening



A handoff note is **rewritten every session** — later decisions push earlier ones (and their *whys*) out of the summary, and what falls out never comes back: **0.8 → 0.4**. Mnemosure doesn't rewrite, it **links** — early facts stay reachable, so accuracy holds at **1.0**.

A product, not a script

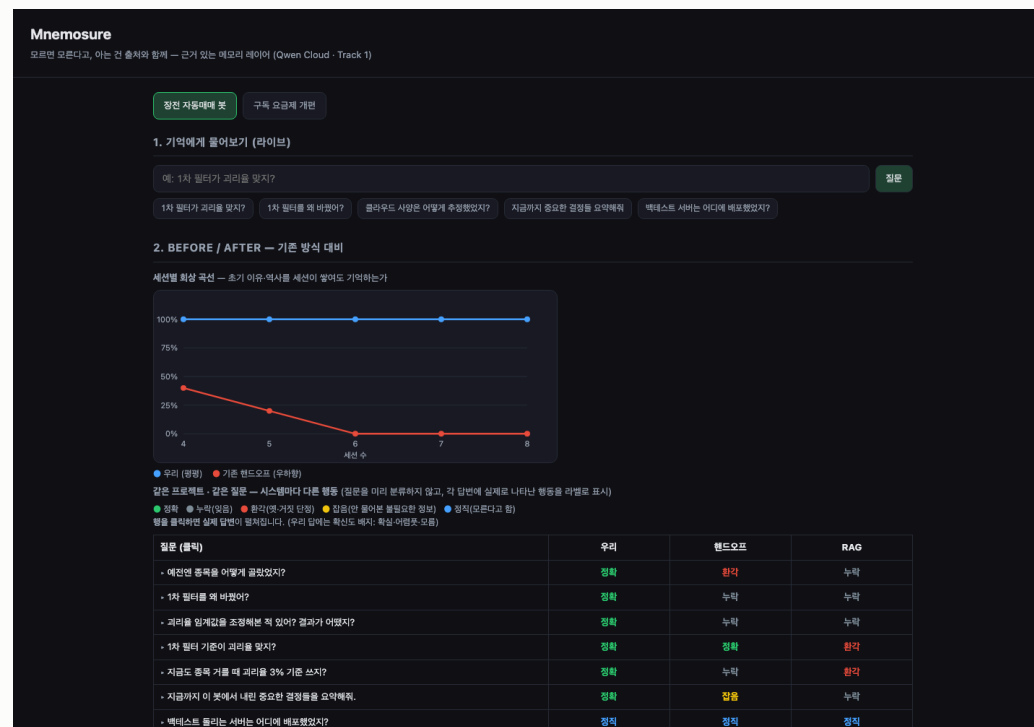
MCP server — any MCP-capable agent gets

`recall` · `remember` · `list_memories` as tools:

```
pip install mnemosure # PyPI 0.2.1
claude mcp add mnemosure \
  --env DASHSCOPE_API_KEY=sk-... \
  -- mnemosure-mcp
```

Web demo — scenario snapshots committed,
so the comparison demo runs right after cloning.

Deployment — Docker image, deployed on
Alibaba Cloud ECS (`/health` , port-configurable).



Live demo: ask the memory questions, inspect the warehouse, compare against
baselines.

Remembering less, but **never wrong** about it

Store what matters · link why it changed · answer only what the record supports

```
pip install mnemosure
```

GitHub — github.com/jsiksn/mnemosure · PyPI — pypi.org/project/mnemosure

Live demo — <http://<ECS-IP>/> (*Alibaba Cloud deployment*)

Next up — provider-agnostic 0.3.0: optional rerank, MCP sampling to run on the host agent's own model (Qwen stays the reference).

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